实验记录_20260131_验证提高神经网络的层数能增宽网络椭圆区域

实验名称：验证提高神经网络的层数能增宽网络椭圆区域

实验时间：20260131

colab链接：https://colab.research.google.com/drive/1h_D9BcOPa4f7-QwbeylVjpghMzo4XeFk?usp=sharing

实验结论：对于beta=0.01的目标函数来说，随着mlp的layers增大，奇点距实数轴的距离先减小后增大，拟合误差同步减小增大

实验目的

实验记录_20260127_目标伯恩斯坦椭圆区域宽度越窄，拟合效果会不会越差 已经证实了网络伯恩斯坦椭圆区域比目标伯恩斯坦椭圆区域大时，拟合效果会变差。
那么我们希望能找到方法来减小网络伯恩斯坦椭圆区域：

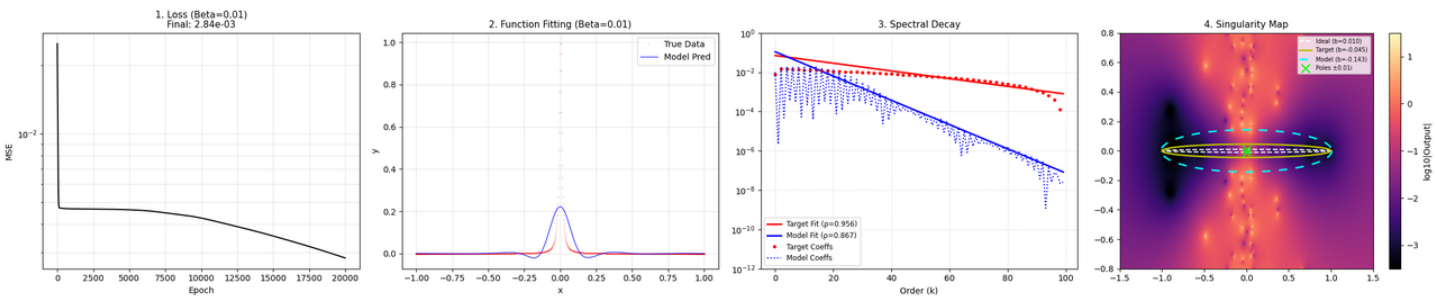
这个实验测试的方法是：**提高mlp神经网络层数**
具体而言是验证在beta为0.01的目标椭圆拟合任务中，将两层隐藏层mlp增大至三层、四层，观测此时拟合效果会变好，且存在一些奇点进入到目标椭圆区域内（可以ylim设小一点）。

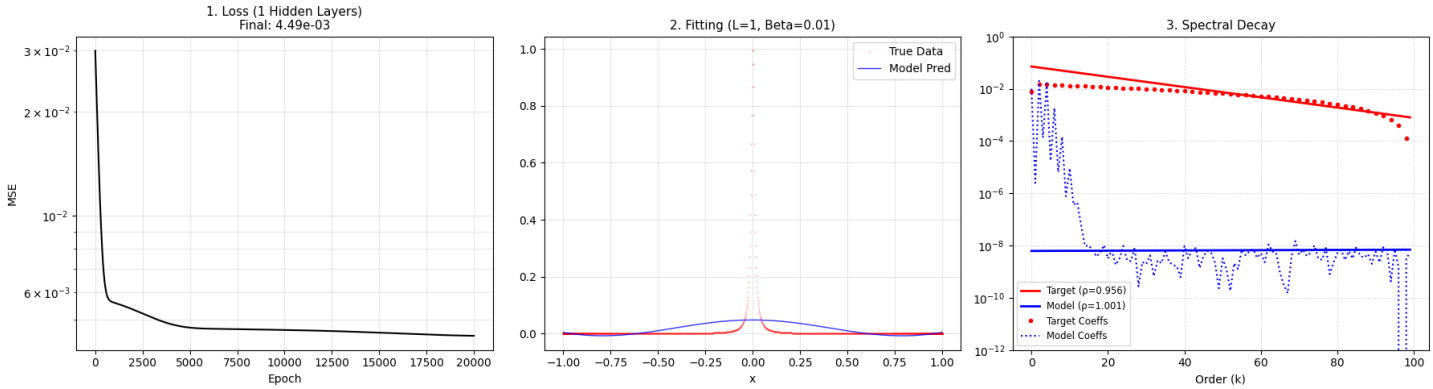
实验过程

先设置一个基线模型：

参考 实验记录_20260127_目标伯恩斯坦椭圆区域宽度越窄，拟合效果会不会越差 的模型架构

基线参数 β 0.01，神经元个数128，学习率1e-5，训练轮次20000epochs，一层隐藏层。



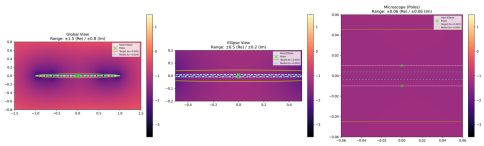
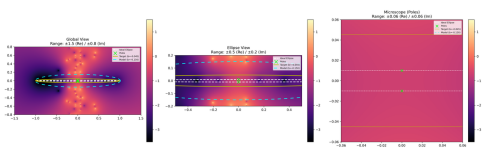
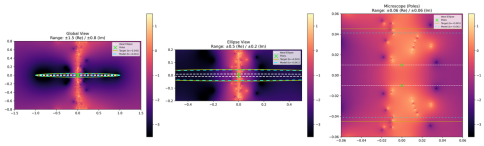
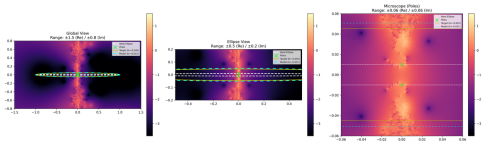


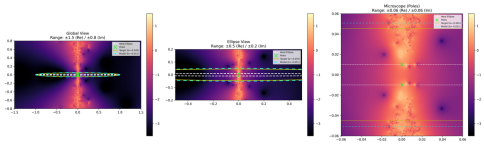
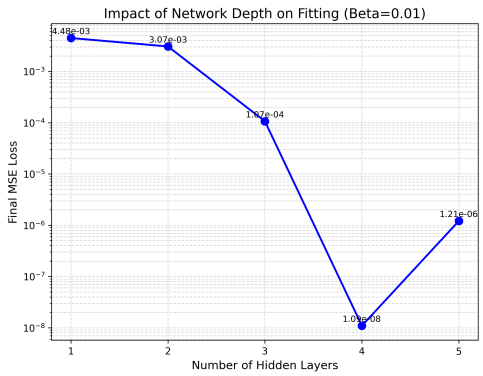
进行一些改变：

增大隐藏层至三层、四层。减小ylim为【-0.2,0.2】

把之前的beta vs loss的图改成 layers vs loss的图

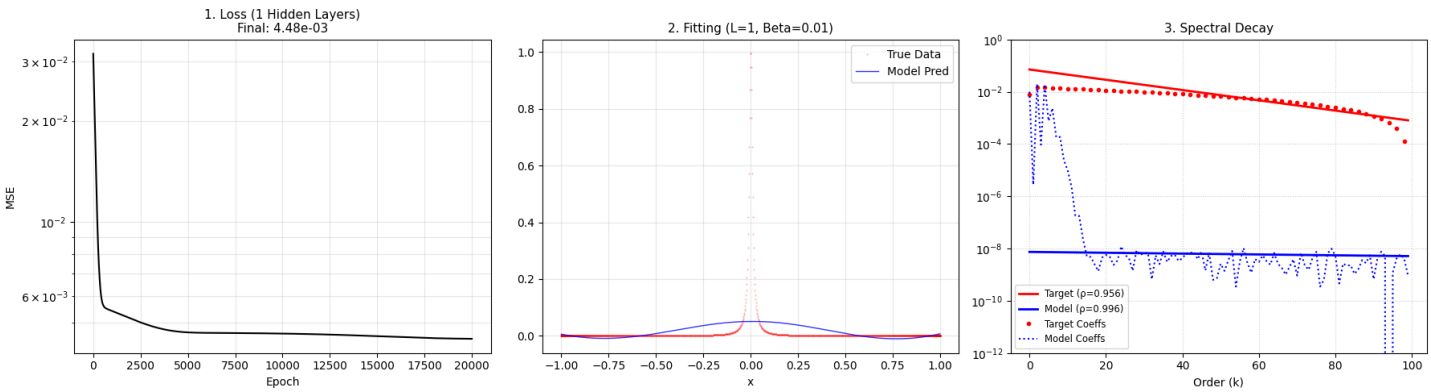
实验组

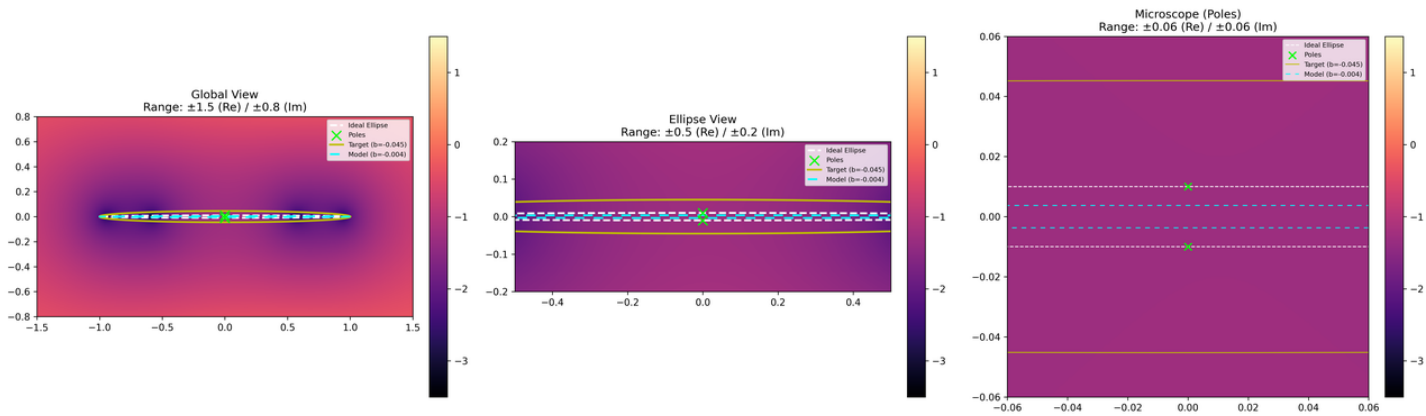
序号	beta	神经元个数	训练轮次	层数	实验结果	实验图
1	0.01	128	20000	一层	此时没有产生奇点 拟合误差为4.48e-3	
2				两层	奇点开始产生，且最近的奇点大概在0.15i处 拟合误差为3.07e-3	
3				三层	最近的奇点在0.013i处 拟合误差为1.07e-4	
4				4	奇点开始进入目标伯恩斯坦椭圆内，	

					最近的奇点在0.007i处 拟合误差为1.09e-8	
5				5	最近的奇点在0.008i处 拟合误差为1.21e-6 Layers vs mse loss的图如右，随着layers增大，奇点距实数轴的距离逐渐减小，且拟合误差也逐渐减小。	 

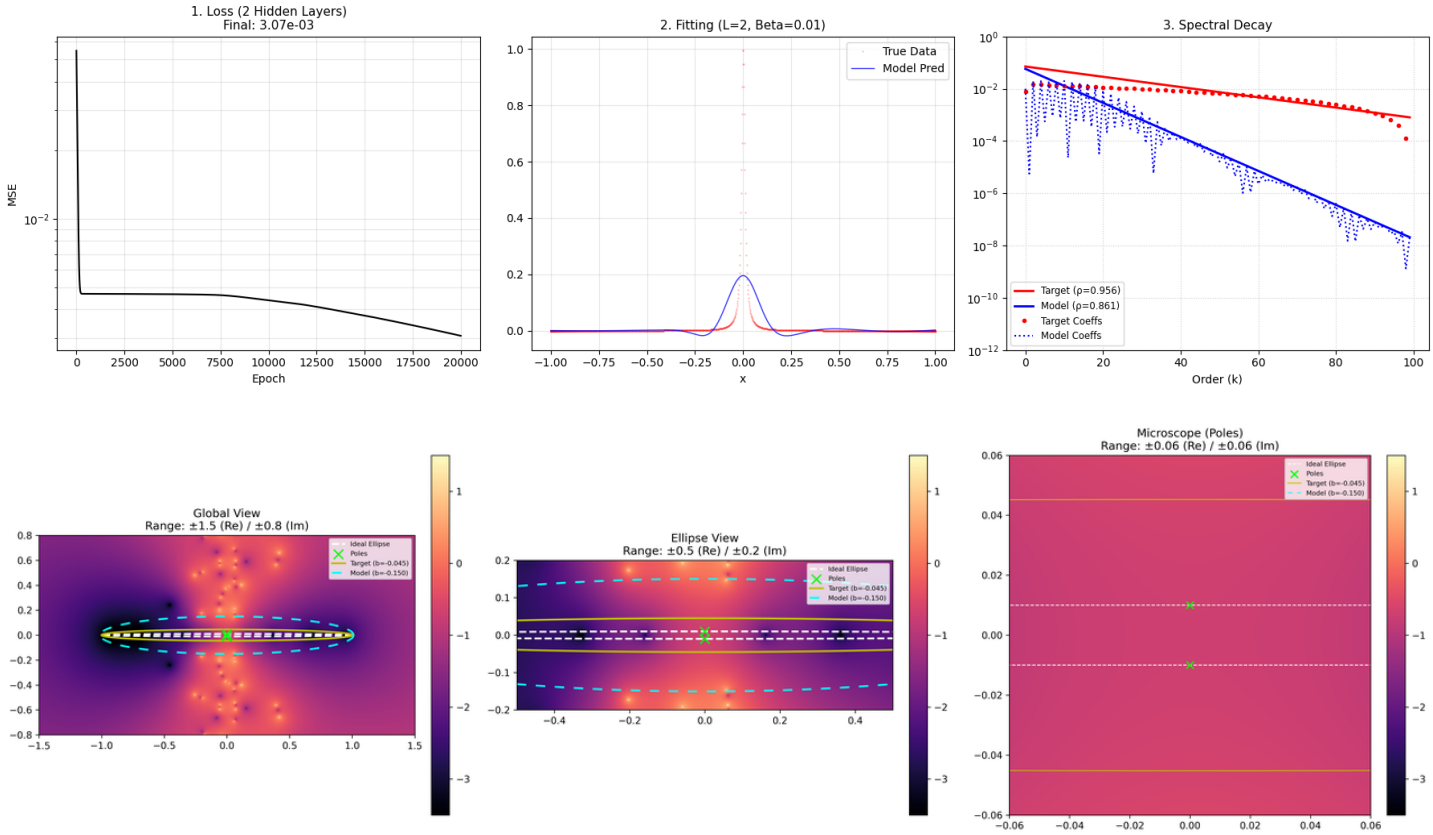
实验图

1layer

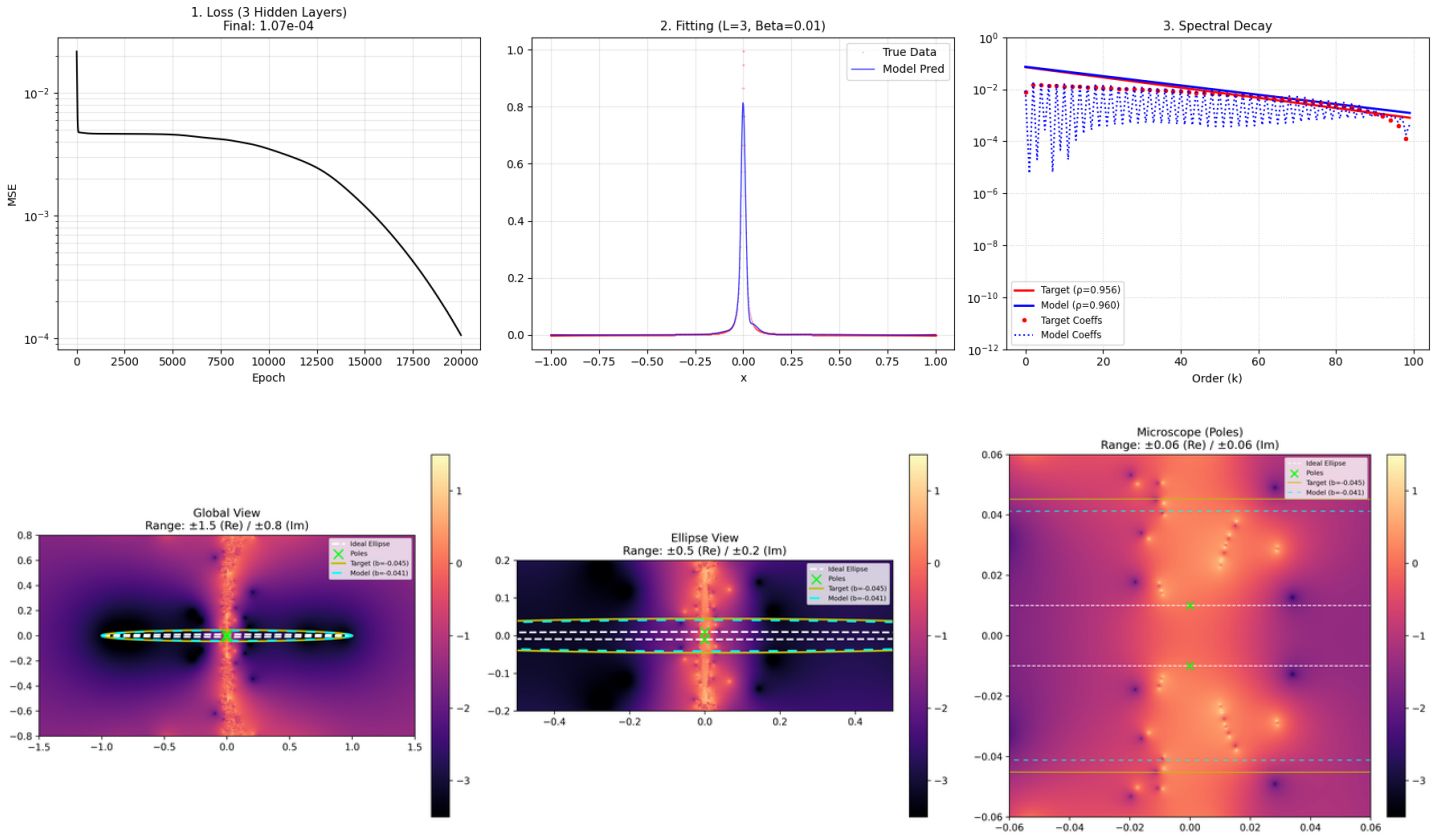




2layer

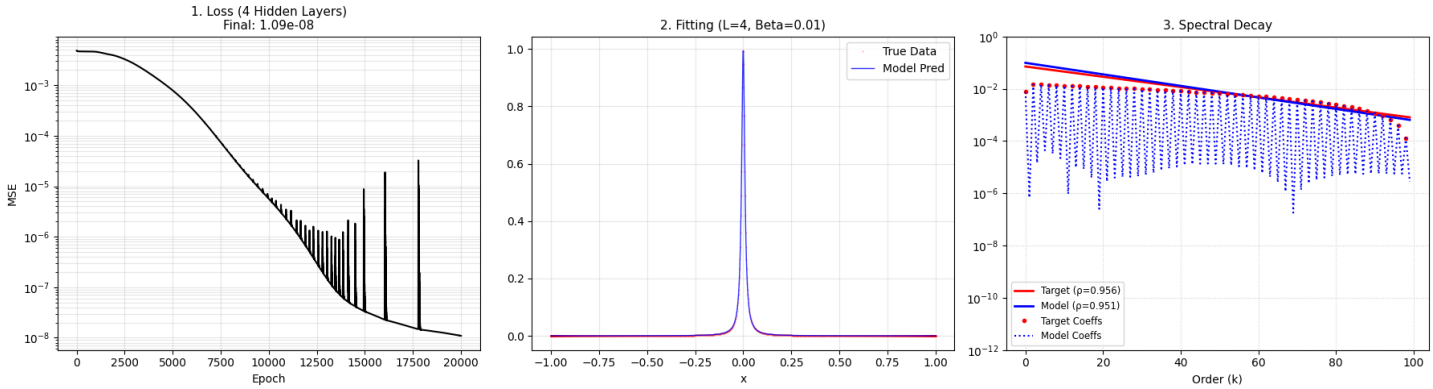


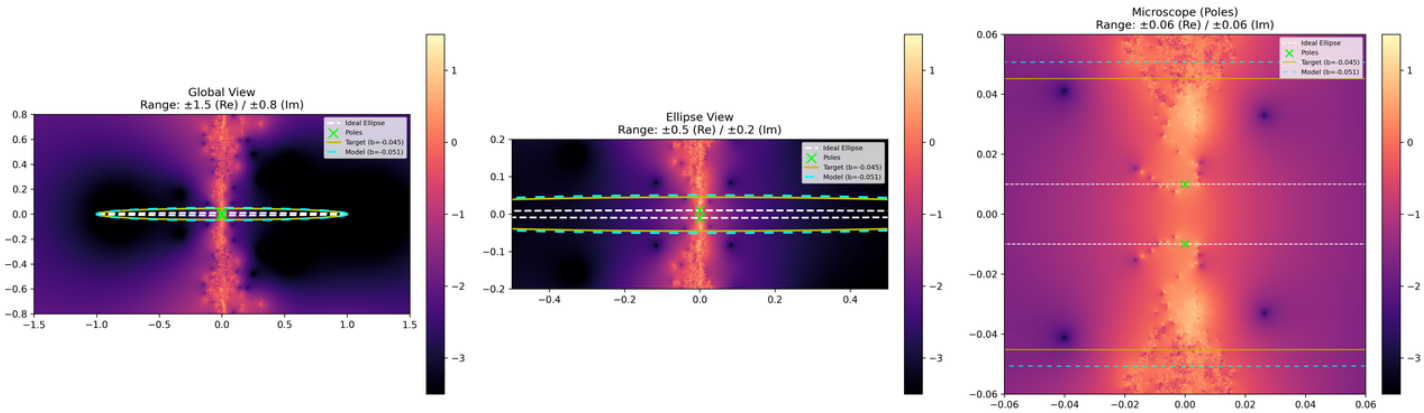
3layer



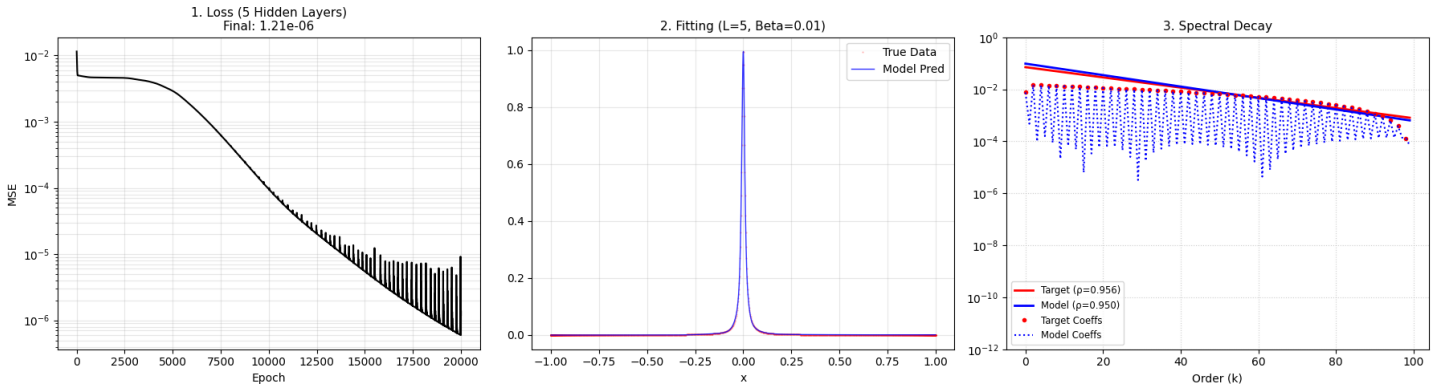
这个图很好看喔

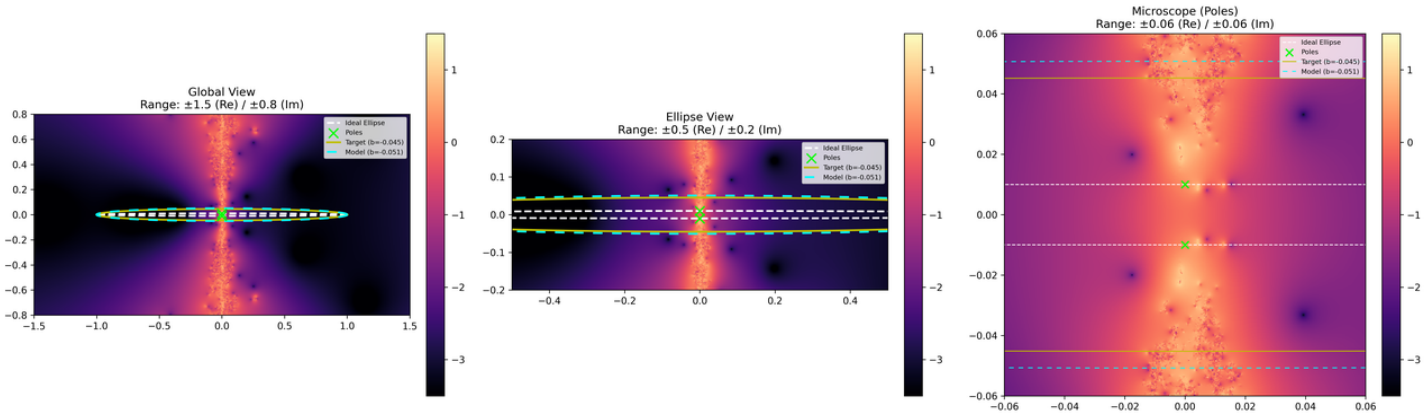
4layer

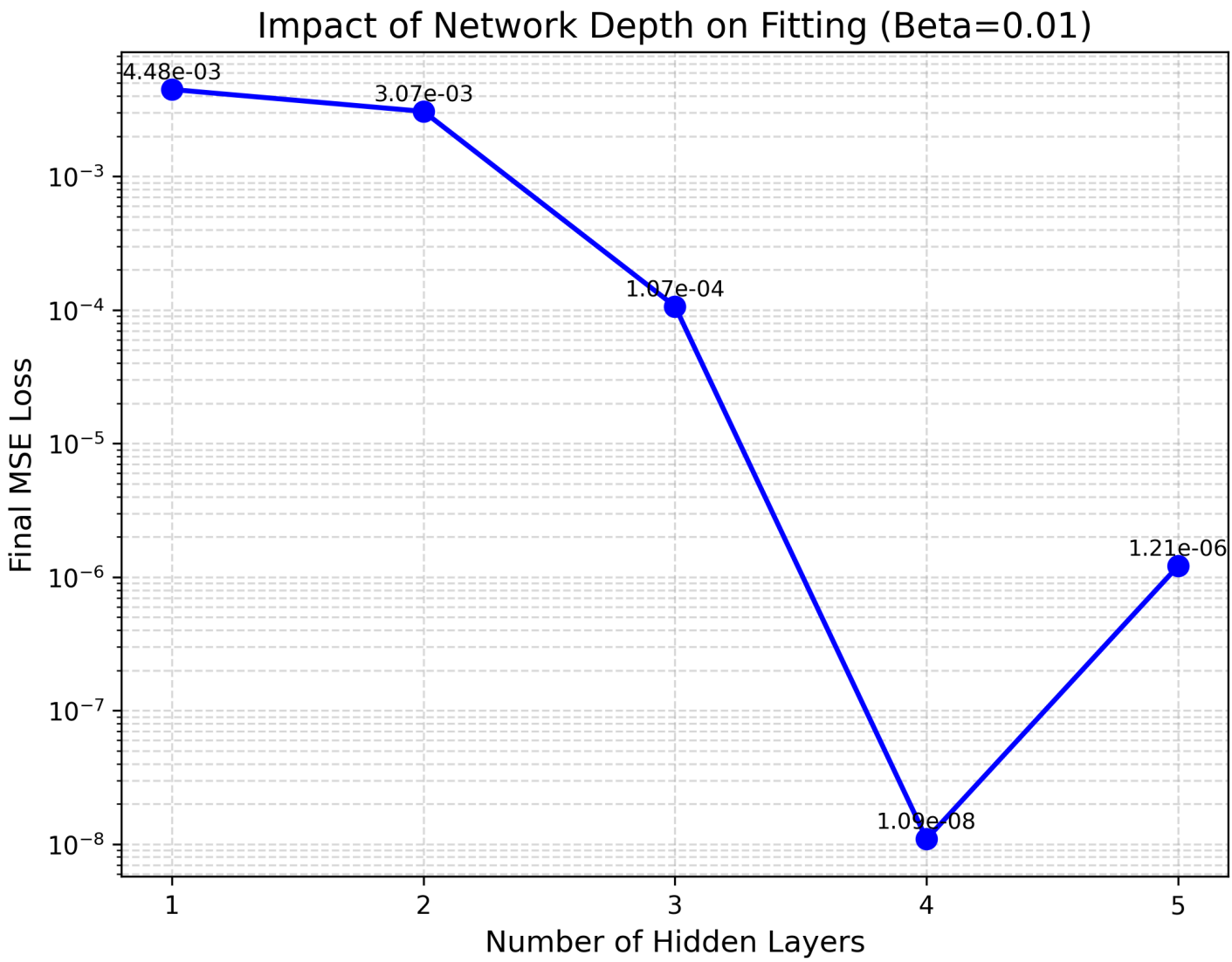

1. Loss (4 Hidden Layers)
Final: 1.09e-08
MSE
Epoch
2. Fitting (L=4, Beta=0.01)
True Data
Model Pred
x
3. Spectral Decay
Target (ρ=0.956)
Model (ρ=0.951)
Target Coeffs
Model Coeffs
Order (k)


Global View
Range: ±1.5 (Re) / ±0.8 (Im)
Ellipse View
Range: ±0.5 (Re) / ±0.2 (Im)
Microscope (Poles)
Range: ±0.06 (Re) / ±0.06 (Im)

5layer


1. Loss (5 Hidden Layers)
Final: 1.21e-06
MSE
Epoch
2. Fitting (L=5, Beta=0.01)
True Data
Model Pred
x
3. Spectral Decay
Target (ρ=0.956)
Model (ρ=0.950)
Target Coeffs
Model Coeffs
Order (k)


Global View
Range: ±1.5 (Re) / ±0.8 (Im)
Ellipse View
Range: ±0.5 (Re) / ±0.2 (Im)
Microscope (Poles)
Range: ±0.06 (Re) / ±0.06 (Im)



实验结论

对于beta=0.01的目标函数来说，随着mlp的layers增大，奇点距实数轴的距离先减小后增大，拟合误差同步减小增大